

A Primer on Pricing Interest Rate Swaps

From first principles to modern multi-curve valuation

July 2026

Contents

A Primer on Pricing Interest Rate Swaps	1
1. What an Interest Rate Swap Is	1
2. The Time Value of Money: Discount Factors, Zero Rates and Forward Rates	2
3. Market Conventions: Day Counts, Schedules and Business Days	3
4. The Core Pricing Principle	4
5. Valuing the Fixed Leg	4
6. Valuing the Floating Leg	5
7. The Par Swap Rate	6
8. The Classical Single-Curve World (and why it broke)	6
9. The Modern Multi-Curve Framework	7
10. Building the Discount Curve: Bootstrapping OIS	8
11. Building the Projection Curve	9
12. Full Worked Example: Pricing a 2-Year Swap	11
13. Marking an Existing (Off-Market) Swap to Market	12
14. Risk: DV01, PV01 and the Annuity	12
15. Practical Complications	13
16. Summary of Key Equations	13

A Primer on Pricing Interest Rate Swaps

From first principles to modern multi-curve valuation

1. What an Interest Rate Swap Is

An **interest rate swap (IRS)** is a bilateral contract in which two parties agree to exchange interest payments on an agreed **notional principal** over a fixed schedule of dates. The notional itself is never exchanged — it only determines the size of the interest payments.

In the standard (“plain vanilla”) swap:

- One party pays a **fixed rate** K , agreed at inception, on every fixed-leg payment date.
- The other party pays a **floating rate**, which resets periodically according to a published reference index (historically LIBOR; today typically an overnight risk-free rate such as SOFR in USD, SONIA in GBP or €STR in EUR, either compounded over the period or as a term rate).

The party who pays fixed and receives floating is the **payer** (they are “paying the swap”). The party who receives fixed and pays floating is the **receiver**.

Why swaps exist. Swaps let institutions transform the interest-rate profile of assets or liabilities without touching the underlying instruments. A company with floating-rate debt can pay fixed on a swap to lock in its funding cost. A pension fund can receive fixed to hedge long-dated liabilities. Traders use swaps to express views on the level and shape of the yield curve. The IRS market is the largest derivatives market in the world by notional outstanding.

A cash flow picture. Consider a 2-year swap, notional 100m, annual fixed payments at $K = 3.15\%$, quarterly floating payments:

Fixed leg (pay): Year 1: -3.15m Year 2: -3.15m
 Floating leg (recv): Q1: +L1 Q2: +L2 Q3: +L3 Q4: +L4 ... Q8: +L8

where each L_j is the floating index observed for that quarter times the quarter's accrual fraction times notional. On dates where both legs pay, only the net amount typically changes hands.

What "pricing" means. There are two closely related tasks:

1. **At inception:** find the fixed rate S (the **par swap rate**) such that the swap has zero value to both parties. No money changes hands upfront; the fixed rate is the price.
2. **After inception:** compute the **mark-to-market value** of an existing swap whose fixed coupon K is no longer the current par rate. This value can be positive or negative and drives collateral (margin) flows.

Both tasks require exactly two ingredients: a way to **discount** future cash flows, and a way to **forecast** the unknown floating payments. Everything in this primer builds those two ingredients and then combines them.

2. The Time Value of Money: Discount Factors, Zero Rates and Forward Rates

2.1 Discount factors

The **discount factor** $d(t)$ is the value today of receiving 1 unit of currency at future time t with certainty. It is the most fundamental object in fixed income:

$$\text{PV of cash flow } C \text{ at time } t = C \times d(t)$$

Properties: $d(0) = 1$, and in normal (positive-rate) environments $d(t)$ is decreasing in t , with $0 < d(t) \leq 1$.

2.2 Zero rates

The **zero rate** (spot rate) $z(t)$ is just a way of quoting the discount factor as an interest rate. With continuous compounding:

$$d(t) = e^{-z(t)t} \quad \Longleftrightarrow \quad z(t) = -\frac{\ln d(t)}{t}$$

With annual compounding: $d(t) = (1 + z(t))^{-t}$. The compounding convention is a choice of units; the discount factor is the underlying object.

Example. If the 2-year continuously compounded zero rate is 3.20%, then $d(2) = e^{-0.032 \times 2} = 0.93807$. Receiving 100 in two years is worth 93.807 today.

2.3 Forward rates

A **forward rate** $F(t_1, t_2)$ is the interest rate for the future period $[t_1, t_2]$ that can be locked in today. No-arbitrage forces it to be consistent with the discount curve: investing to t_2 directly must equal investing to t_1 and rolling at the forward rate.

For a simple (money-market style) forward rate with accrual fraction $\tau =$ year fraction of $[t_1, t_2]$:

$$(1 + \tau F(t_1, t_2)) = \frac{d(t_1)}{d(t_2)} \quad \Rightarrow \quad \boxed{F(t_1, t_2) = \frac{1}{\tau} \left(\frac{d(t_1)}{d(t_2)} - 1 \right)}$$

Example. Suppose $d(0.75) = 0.97655$ and $d(1.00) = 0.96900$, and the period $[0.75, 1.00]$ has $\tau = 0.25$. The implied forward rate is

$$F = \frac{1}{0.25} \left(\frac{0.97655}{0.96900} - 1 \right) = \frac{1}{0.25} (1.007792 - 1) = 3.117\%$$

This forward rate is the market's arbitrage-free "prediction" of the floating rate for that period — and, crucially, it is the correct rate to use when valuing an uncertain floating payment, because a floating payment can be replicated (hedged) using instruments priced off the curve. We do not need to know where rates will actually go; we only need the replication cost.

2.4 The expectation interpretation (brief)

Under the pricing measure associated with the discount curve (the t_2 -forward measure), the forward rate equals the expected value of the future floating fixing:

$$F(t_1, t_2) = \mathbb{E}^{t_2} [L(t_1, t_2)]$$

For linear products like vanilla swaps this means we can value floating cash flows by simply substituting forward rates — no volatility model needed. (Options on rates, e.g. caps and swaptions, do require a volatility model; that is beyond this primer.)

3. Market Conventions: Day Counts, Schedules and Business Days

Real swaps are defined by term sheets full of conventions. Getting these wrong produces pricing errors far larger than any modelling subtlety.

3.1 Day count conventions

The **accrual fraction** τ_i converts a calendar period into a fraction of a year. Common conventions:

Convention	Rule	Typical use
Act/360	actual days $\div$ 360	USD/EUR floating legs, money markets
Act/365 (Fixed)	actual days $\div$ 365	GBP legs (SONIA), some fixed legs
30/360	months count as 30 days, year as 360	Older USD/EUR fixed legs
Act/Act	actual days $\div$ actual days in year	Government bonds

Example. A floating period from 15 Jan to 15 Apr (90 actual days): under Act/360, $\tau = 90/360 = 0.25$; under Act/365, $\tau = 90/365 = 0.24658$. On a 100m notional at 3%, that difference is worth about 2,850 — per period.

3.2 Payment schedules

Each leg has its own frequency. Typical vanilla conventions:

- **USD SOFR swaps:** fixed annual Act/360 vs SOFR compounded-in-arrears annual Act/360 (cleared standard), or legacy fixed semi-annual 30/360 vs 3M floating Act/360.
- **EUR swaps:** fixed annual 30/360 vs 6M EURIBOR Act/360 (or €STR).
- **GBP swaps:** fixed annual Act/365 vs SONIA compounded Act/365.

The fixed and floating legs generally have **different frequencies and different day counts**. The pricing formulas below keep separate accrual fractions τ_i^{fx} and τ_j^{fl} for this reason.

3.3 Business day adjustment

Payment dates falling on weekends/holidays roll according to a rule, most commonly **Modified Following**: move to the next business day unless that crosses into a new month, in which case move back. Adjusted dates change the accrual fractions, which is why production systems generate schedules date-by-date.

For clarity, the worked examples in this primer use round year fractions ($\tau = 1$ or 0.25) — but remember that real systems compute every τ from actual dates.

4. The Core Pricing Principle

The value of a swap is the value of its fixed leg minus the value of its floating leg (or vice versa depending on direction). Each leg is just a strip of cash flows, and each cash flow is valued as:

$$PV = (\text{forecast cash flow}) \times (\text{discount factor})$$

- Fixed cash flows are **known**: coupon $\times$ accrual $\times$ notional.
- Floating cash flows are **forecast** using forward rates read off a curve.
- Everything is then **discounted** back to today.

To the **fixed-rate payer** (pay fixed, receive float):

$$V_{\text{payer}} = PV_{\text{float}} - PV_{\text{fixed}}, \quad V_{\text{receiver}} = -V_{\text{payer}}$$

The entire art of swap pricing is constructing the curves that supply the forwards and the discount factors. Sections 8–11 cover that. First, the leg formulas.

5. Valuing the Fixed Leg

The fixed leg pays coupon K on notional N at dates $t_1 < t_2 < \dots < t_n$, with accrual fractions τ_i^{fx} :

$$PV_{\text{fixed}} = N K \sum_{i=1}^n \tau_i^{fx} d(t_i)$$

The sum

$$A = \sum_{i=1}^n \tau_i^{fx} d(t_i)$$

is called the **annuity** (or PV01 basis, or “level”). It is the present value of receiving 1 unit of rate on the fixed schedule, and it appears everywhere: in the par rate, in risk, in swaption pricing.

Example. 2-year swap, annual fixed leg, $K = 3.15\%$, notional 100m, with $d(1) = 0.97087$ and $d(2) = 0.93889$, and $\tau_1 = \tau_2 = 1$:

$$A = 1 \times 0.97087 + 1 \times 0.93889 = 1.90976$$

$$PV_{\text{fixed}} = 100\text{m} \times 0.0315 \times 1.90976 = 6,015,700$$

6. Valuing the Floating Leg

The floating leg pays, at each date t_j , the index fixing for the period $[t_{j-1}, t_j]$ times the accrual τ_j^{fl} times notional. Each unknown future fixing is replaced by its forward rate F_j read from the **projection curve**, then discounted:

$$PV_{\text{float}} = N \sum_{j=1}^m \tau_j^{fl} F_j d(t_j)$$

Two refinements:

1. **The current period is (partly) known.** If the swap is already running, the fixing for the current period has been set (term rates) or is accruing from published daily fixings (compounded overnight rates). Use the known/accrued value, not a forward.
2. **Which curve supplies F_j ?** In the modern framework, a projection curve specific to the index — this is the heart of Section 9.

6.1 The classic telescoping result

Suppose (as in the pre-2008 world, or for OIS swaps today) that the floating index and the discount curve are **the same curve**. Then each forward is $F_j = \frac{1}{\tau_j} \left(\frac{d(t_{j-1})}{d(t_j)} - 1 \right)$, and:

$$PV_{\text{float}} = N \sum_{j=1}^m \tau_j F_j d(t_j) = N \sum_{j=1}^m (d(t_{j-1}) - d(t_j)) = N(d(t_0) - d(t_m))$$

Every intermediate discount factor cancels — the sum **telescopes**. If the swap starts today ($t_0 = 0$, $d(t_0) = 1$):

$$PV_{\text{float}} = N(1 - d(t_m))$$

Intuition: a floating-rate note that pays the index and returns principal at maturity is always worth par at reset dates. The floating leg is that note minus the principal repayment: $N - N d(t_m)$.

This beautiful shortcut is **only valid when projection curve = discount curve**. It fails in the general multi-curve setting — but it survives for OIS swaps, which is exactly what makes OIS bootstrapping clean (Section 10).

7. The Par Swap Rate

The par rate S is the fixed rate making the swap worth zero at inception:

$$PV_{\text{fixed}} = PV_{\text{float}} \implies S \sum_{i=1}^n \tau_i^{fx} d(t_i) = \sum_{j=1}^m \tau_j^{fl} F_j d(t_j)$$

$$S = \frac{\sum_{j=1}^m \tau_j^{fl} F_j d(t_j)}{\sum_{i=1}^n \tau_i^{fx} d(t_i)} = \frac{PV_{\text{float per unit notional}}}{A}$$

The par rate is the floating-leg PV divided by the annuity. It is a weighted average of the forward rates, with weights $\tau_j d(t_j)$ — which is why the swap curve smooths through the forward curve.

In the special single-curve case where telescoping applies:

$$S = \frac{1 - d(t_n)}{\sum_{i=1}^n \tau_i d(t_i)}$$

Example (single-curve shortcut). With $d(1) = 0.97087$, $d(2) = 0.93889$, annual fixed leg:

$$S = \frac{1 - 0.93889}{1.90976} = \frac{0.06111}{1.90976} = 3.200\%$$

8. The Classical Single-Curve World (and why it broke)

Before 2008, the market used **one curve** — the LIBOR swap curve — for everything: forecasting LIBOR fixings *and* discounting cash flows. This was internally consistent because banks assumed they funded at LIBOR, LIBOR was treated as (nearly) risk-free, and basis spreads between different floating tenors (e.g. 3M vs 6M LIBOR) were a fraction of a basis point.

The 2008 break. During the financial crisis:

- The **LIBOR-OIS spread** — the gap between 3M LIBOR and the expected overnight rate over the same period — exploded from single digits to over 350bp. LIBOR was revealed to contain substantial bank credit and liquidity premia; it was not a risk-free rate.
- **Tenor basis** blew out: 3M vs 6M LIBOR basis swaps went from ~0bp to tens of basis points. One curve cannot reprice instruments on two tenors that trade at a persistent spread.
- **Collateralization became universal.** Cleared and CSA-collateralized swaps earn/pay the overnight rate on posted variation margin. By a replication argument, a collateralized cash flow should be discounted at the **collateral rate** — the OIS rate — not at LIBOR.

The consequences for pricing:

1. **Discount at OIS** (the collateral/overnight rate curve), because that is the true funding rate of a collateralized trade.
2. **Project each floating index off its own curve**, calibrated to instruments referencing that index, so that observed basis spreads are repriced exactly.

The subsequent **LIBOR transition** (completed for most currencies by 2021–2023) replaced LIBOR with overnight risk-free rates (SOFR, SONIA, €STR) as the floating index itself. For a standard SOFR-vs-fixed swap, the projection and discount curves coincide again — a partial return to single-curve simplicity — but the multi-curve framework remains essential wherever the floating index differs from the collateral rate (EURIBOR swaps discounted at €STR, cross-currency trades, term-rate legs, legacy books).

9. The Modern Multi-Curve Framework

The modern setup, stated compactly:

Discount every cash flow on the curve of the collateral rate. Forecast every floating fixing on a curve calibrated to that specific index.

Formally, for a collateralized swap:

$$V = N \left[\sum_j \tau_j^{fl} F_j^{\text{proj}} d^{\text{OIS}}(t_j) - K \sum_i \tau_i^{fx} d^{\text{OIS}}(t_i) \right] \quad (\text{payer})$$

where:

- $d^{\text{OIS}}(t)$: discount factors from the OIS curve of the **collateral currency/rate** (SOFR for USD-collateralized, €STR for EUR, SONIA for GBP).
- F_j^{proj} : forward rates from the **projection curve** of the swap’s floating index. There is a separate projection curve for each index and tenor: 3M EURIBOR, 6M EURIBOR, term SOFR, etc. The gaps between these curves and the OIS curve are the **basis spreads** the market trades directly via basis swaps.

The projection curve has its own “pseudo discount factors” $P(t)$, used **only** to generate forwards:

$$F_j^{\text{proj}} = \frac{1}{\tau_j} \left(\frac{P(t_{j-1})}{P(t_j)} - 1 \right)$$

$P(t)$ is never used to discount. In general $P(t) \neq d^{\text{OIS}}(t)$, and the telescoping shortcut of Section 6.1 **does not apply** — the floating leg must be summed forward-by-forward.

Curve construction is sequential:

1. Build $d^{\text{OIS}}(t)$ from OIS quotes (Section 10). This can be done standalone because OIS instruments’ floating legs live on the same curve that discounts them.
2. Build each projection curve $P(t)$ from instruments referencing that index, **discounting their cash flows with the already-built d^{OIS}** (Section 11).

(Where the market quotes basis swaps or where collateral is in a foreign currency, curves may need to be solved jointly/globally rather than sequentially — noted in Section 15 — but the sequential picture is the right mental model.)

10. Building the Discount Curve: Bootstrapping OIS

10.1 The instruments

An **overnight index swap (OIS)** exchanges a fixed rate against the **geometrically compounded** overnight rate over each period:

$$R_{\text{cmp}}(t_{j-1}, t_j) = \frac{1}{\tau_j} \left[\prod_{k \in \text{period}} (1 + \delta_k r_k) - 1 \right]$$

where r_k are the daily overnight fixings (SOFR, €STR, SONIA) and δ_k the daily accruals ($\approx 1/360$ or $1/365$). Market quotes exist from 1 week out to 30, 40, even 50 years. Short OIS ($\leq 1Y$) typically have a single payment at maturity; longer OIS pay annually.

10.2 Why OIS bootstraps cleanly

Because the compounded overnight rate **is** the growth rate of the discount curve itself, the OIS floating leg telescopes (Section 6.1). The par condition for an OIS of maturity t_n with annual payments and quoted rate S_n^{OIS} is:

$$S_n^{\text{OIS}} \sum_{i=1}^n \tau_i d(t_i) = 1 - d(t_n)$$

One equation, and — if all shorter pillars are already solved — one unknown, $d(t_n)$:

$$d(t_n) = \frac{1 - S_n^{\text{OIS}} \sum_{i=1}^{n-1} \tau_i d(t_i)}{1 + S_n^{\text{OIS}} \tau_n}$$

Bootstrapping = applying this recursively from the shortest maturity to the longest. Each market quote pins exactly one new discount factor.

10.3 Worked bootstrap

Market OIS quotes (annual payments, $\tau = 1$ for simplicity):

Tenor	OIS rate
1Y	3.00%
2Y	3.20%
3Y	3.35%

1Y (single payment):

$$d(1) = \frac{1}{1 + 0.0300 \times 1} = 0.970874$$

2Y:

$$d(2) = \frac{1 - 0.0320 \times d(1)}{1 + 0.0320} = \frac{1 - 0.0320 \times 0.970874}{1.0320} = \frac{0.968932}{1.0320} = 0.938887$$

3Y:

$$d(3) = \frac{1 - 0.0335 \times (d(1) + d(2))}{1 + 0.0335} = \frac{1 - 0.0335 \times 1.909761}{1.0335} = \frac{0.936023}{1.0335} = 0.905683$$

Check: implied zero rates $z(t) = -\ln d(t)/t$ give 2.956%, 3.153%, 3.302% — an upward-sloping curve consistent with the quotes.

10.4 Interpolation

Market quotes exist only at pillar dates; the swap being priced needs discount factors at arbitrary dates. The standard choice is **linear interpolation in $\ln d(t)$** (equivalently, piecewise-flat instantaneous forward rates):

$$\ln d(t) = \ln d(t_a) + \frac{t - t_a}{t_b - t_a} [\ln d(t_b) - \ln d(t_a)], \quad t_a \leq t \leq t_b$$

Example. $d(1.5)$ from the curve above: $\ln d(1.5) = \frac{1}{2} [\ln(0.970874) + \ln(0.938887)] = \frac{1}{2} [-0.029559 - 0.063060] = -0.046310$, so $d(1.5) = 0.954746$.

More sophisticated schemes (monotone-convex, tension splines) produce smoother forward curves; the trade-off is between smoothness and locality (how far a quote perturbation propagates). In every case the bootstrap logic is the same — only the in-between behaviour differs. Where interpolation makes intermediate cash-flow dates depend on later pillars, the “bootstrap” becomes a small root-finding problem per pillar (or a global solve), but conceptually nothing changes.

11. Building the Projection Curve

Now suppose we must price swaps whose floating leg references an index that is **not** the compounded overnight rate — e.g. 6M EURIBOR or a term rate. We need a curve of forwards for that index.

11.1 The instruments

Ordered from short to long maturity:

1. **The spot fixing / deposit** for the index tenor (e.g. today’s 6M EURIBOR fixing) — pins the first point.
2. **FRAs or futures** on the index — pin individual forward periods over the next 1–2 years. (Futures require a small **convexity adjustment** because of daily margining; noted, not derived, here.)
3. **Par swaps** vs that index — pin the curve from ~2Y out to the long end.

11.2 The calibration condition

Each calibration swap of maturity t_n and quoted rate S_n must be repriced to zero **using OIS discounting**:

$$S_n \sum_{i=1}^n \tau_i^{fx} d^{\text{OIS}}(t_i) = \sum_{j=1}^m \tau_j^{fl} F_j d^{\text{OIS}}(t_j)$$

The left side is fully known (quote + OIS curve). On the right side, forwards covered by shorter instruments are already pinned; the newest instrument determines the remaining one(s). Solving sequentially recovers the projection curve's pseudo discount factors $P(t)$ via $F_j = \frac{1}{\tau_j} \left(\frac{P(t_{j-1})}{P(t_j)} - 1 \right)$.

Note the structural point: **the telescoping trick is unavailable** because F_j comes from P while discounting uses d^{OIS} . Every forward must be handled explicitly.

11.3 Worked projection bootstrap

Stay with the OIS curve of Section 10.3. Suppose the floating index is an annual term rate (to keep the arithmetic compact — the logic is identical for quarterly or semi-annual), and the market quotes fixed-vs-index par swaps:

Tenor	Par swap rate vs index
1Y	3.30%
2Y	3.50%

Note both quotes sit **above** the corresponding OIS rates (3.00%, 3.20%): the index embeds a credit/liquidity premium over the overnight rate — a 30bp basis.

1Y swap (one fixed payment, one floating payment):

$$0.0330 \times 1 \times d^{\text{OIS}}(1) = F_1 \times 1 \times d^{\text{OIS}}(1) \implies F_1 = 3.300\%$$

2Y swap:

$$0.0350 \times [d^{\text{OIS}}(1) + d^{\text{OIS}}(2)] = F_1 d^{\text{OIS}}(1) + F_2 d^{\text{OIS}}(2)$$

$$0.0350 \times 1.909761 = 0.0330 \times 0.970874 + F_2 \times 0.938887$$

$$0.066842 = 0.032039 + 0.938887 F_2 \implies F_2 = \frac{0.034803}{0.938887} = 3.707\%$$

The projection curve implies the index will fix at 3.300% for year 1 and forwards at 3.707% for year 2. The corresponding pseudo discount factors: $P(1) = 1/(1 + F_1) = 0.968054$; $P(2) = P(1)/(1 + F_2) = 0.933447$. Observe $P(2) = 0.933447 \neq d^{\text{OIS}}(2) = 0.938887$: the wedge between the curves is exactly the basis.

12. Full Worked Example: Pricing a 2-Year Swap

Trade: 2-year payer swap (pay fixed, receive the index), notional **100,000,000**, annual payments both legs, $\tau = 1$ throughout. Curves as built above:

t	$d^{\text{OIS}}(t)$	Forward F_t (projection)
1Y	0.970874	3.300%
2Y	0.938887	3.707%

Step 1 — Par rate

Annuity:

$$A = d^{\text{OIS}}(1) + d^{\text{OIS}}(2) = 0.970874 + 0.938887 = 1.909761$$

Floating leg PV per unit notional:

$$\begin{aligned} \text{PV}_{\text{float}} &= F_1 d^{\text{OIS}}(1) + F_2 d^{\text{OIS}}(2) = 0.0330 \times 0.970874 + 0.03707 \times 0.938887 \\ &= 0.032039 + 0.034805 = 0.066844 \end{aligned}$$

Par rate:

$$S = \frac{0.066844}{1.909761} = 3.500\%$$

(Reassuringly, this recovers the 2Y market quote — the curves reprice their own calibration instruments. For a maturity **between** pillars, the same formula runs on interpolated discount factors and forwards.)

Step 2 — Value the trade at an off-market coupon

Suppose instead the swap was struck some time ago at $K = 3.15\%$ and we now mark it (assume, for clean numbers, it has exactly 2 annual periods remaining and no accrued stub).

Fixed leg:

$$\text{PV}_{\text{fixed}} = 100\text{m} \times 0.0315 \times 1.909761 = 6,015,747$$

Floating leg:

$$\text{PV}_{\text{float}} = 100\text{m} \times 0.066844 = 6,684,400$$

Value to the payer:

$$V = 6,684,400 - 6,015,747 = +668,653$$

The payer locked in paying 3.15% when the market now demands 3.50% — the position is in the money.

Step 3 — Sanity check via the annuity shortcut

For a swap with matching schedules, the value equals the rate differential times the annuity:

$$V = N(S - K)A = 100\text{m} \times (0.0350 - 0.0315) \times 1.909761 = 668,416$$

The 237 difference vs Step 2 is rounding in the intermediate figures — the identity $V = N(S - K)A$ is exact. This formula is the fastest way to think about swap MTM: **value = (par rate – coupon) × annuity × notional**.

13. Marking an Existing (Off-Market) Swap to Market

Day-to-day valuation of a live swap follows Section 12 Step 2 with three practical additions:

1. **The current floating period.** Its rate is no longer a forward:
 - *Term-rate legs (e.g. EURIBOR):* the fixing was set at the period start — use it.
 - *Compounded overnight legs (SOFR/SONIA/€STR):* the coupon accrues from published daily fixings; the realized portion is known, the remainder is projected off the curve. The period's cash flow is the compounded blend of both.
2. **Accrued interest.** Between payment dates each leg has accrued a partial coupon; clean vs dirty valuation conventions apply just as in bonds.
3. **Settlement/valuation date.** Discount factors are taken to the valuation date; a spot-lag (e.g. T+2) may apply to the curve construction itself.

The mark drives daily **variation margin**: the losing side posts collateral equal to the change in V , and earns the overnight rate on it — which is precisely the self-consistency that justifies OIS discounting.

14. Risk: DV01, PV01 and the Annuity

Pricing and risk come from the same objects. From $V = N(S - K)A$:

$$\frac{\partial V}{\partial S} = N \times A$$

DV01 (dollar value of a basis point) — the P&L from a 1bp parallel move in the par rate:

$$\text{DV01} \approx N \times A \times 0.0001$$

Example. Our 2Y, 100m swap: $\text{DV01} = 100\text{m} \times 1.909761 \times 0.0001 = 19,098$ per basis point. A payer position gains ~19k if 2Y rates rise 1bp.

In practice desks compute **bucketed risk**: bump each input quote (each pillar of the OIS and projection curves) by 1bp, rebuild the curves, revalue the book. The resulting ladder of sensitivities per pillar is what gets hedged, typically with liquid par swaps and futures. Two subtleties worth knowing exist even if they rarely dominate:

- **Discounting risk vs projection risk** are distinct: a EURIBOR swap has sensitivity both to EURIBOR-curve quotes (through the forwards) and to OIS quotes (through discounting and through the calibration coupling of Section 11).
- **Convexity:** V is not exactly linear in rates because the discount factors themselves move; for large moves the second-order term (gamma) matters, and receivers are systematically long convexity relative to payers at the same DV01.

15. Practical Complications

A production pricing library handles, at minimum:

- **Stub periods.** Trades whose tenor is not a whole number of periods have a short or long first (or last) period, with an interpolated or specially-fixed rate.
- **Payment lags.** Compounded overnight legs usually pay 1–2 business days after the period ends (to allow the final fixings to publish); the discount date must reflect the lag.
- **Futures convexity adjustment.** Daily-margined futures imply forwards that are biased high relative to FRA forwards; a model-based adjustment (increasing with volatility and maturity) is applied before futures enter the curve.
- **Turn-of-year / meeting-date effects.** Overnight rates jump at central bank meeting dates and year-ends; good curve construction places pillars at meeting dates and allows jumps rather than smoothing through them.
- **CSA and collateral optionality.** If collateral can be posted in multiple currencies/assets, the discount curve should reflect the cheapest-to-deliver collateral — a material effect on long-dated trades.
- **Cross-currency basis.** Discounting a trade collateralized in a different currency requires the FX-implied discount curve, built from cross-currency basis swaps.
- **XVA.** For uncollateralized or imperfectly collateralized counterparties, the “risk-free” value above is adjusted for counterparty credit (CVA), own credit (DVA), funding (FVA), capital (KVA) and margin (MVA). These are portfolio-level, model-dependent adjustments layered on top of the machinery in this primer.
- **Global calibration.** Where instruments couple curves (basis swaps, cross-currency swaps), sequential bootstrapping gives way to solving all curves simultaneously as one root-finding problem over all pillar values.

None of these change the core logic: **forecast on the projection curve, discount on the collateral curve, net the legs.**

16. Summary of Key Equations

Object	Formula
Discount factor / zero rate	$d(t) = e^{-z(t)t}$
Simple forward from a curve	$F(t_1, t_2) = \frac{1}{\tau} \left(\frac{d(t_1)}{d(t_2)} - 1 \right)$
Fixed leg PV	$NK \sum_i \tau_i^{fx} d^{\text{OIS}}(t_i) = NKA$
Floating leg PV (multi-curve)	$N \sum_j \tau_j^{fl} F_j^{\text{proj}} d^{\text{OIS}}(t_j)$
Floating leg PV (single-curve only)	$N(1 - d(t_m))$
Par swap rate	$S = \frac{\sum_j \tau_j^{fl} F_j d^{\text{OIS}}(t_j)}{\sum_i \tau_i^{fx} d^{\text{OIS}}(t_i)}$
OIS bootstrap step	$d(t_n) = \frac{1 - S_n^{\text{OIS}} \sum_{i < n} \tau_i d(t_i)}{1 + S_n^{\text{OIS}} \tau_n}$
Projection calibration	$S_n \sum_i \tau_i^{fx} d^{\text{OIS}}(t_i) = \sum_j \tau_j^{fl} F_j d^{\text{OIS}}(t_j)$
Off-market swap value (payer)	$V = N(S - K)A$
DV01	$DV01 \approx N \times A \times 0.0001$

Object	Formula
--------	---------

The pipeline in one paragraph. Collect OIS quotes and bootstrap the discount curve, one pillar per quote, using the telescoping par condition. Collect instruments on the floating index (fixing, FRAs/futures, par swaps) and bootstrap the projection curve, repricing each instrument with OIS discounting. To price any swap: read forwards off the projection curve, multiply by accruals and OIS discount factors to get the floating PV; multiply the coupon by the OIS annuity for the fixed PV; net the legs. The par rate is floating PV over annuity, the mark is $(S - K) \times \text{annuity} \times \text{notional}$, and the DV01 is the annuity in basis-point units.